Lab 33 — Pandas File I/O (Reading and Writing CSV)

Aim

To read data from a CSV file into a Pandas DataFrame, process the data, and write the processed DataFrame to a new CSV file.

Algorithm / Procedure

1. Import the Pandas library.
2. Create a sample student DataFrame and save it as a CSV file.
3. Read the CSV file using pd.read_csv().
4. Create a pass column based on marks.
5. Save the processed DataFrame using to_csv().
6. Display the original and processed data.

Program

```
import pandas as pd

data={'name':['A','B','C','D'],'marks':[78,45,65,38],'class':['I','II','I','II']}

pd.DataFrame(data).to_csv('students.csv',index=False)

df=pd.read_csv('students.csv')

print("Loaded data:\n",df)

df['pass']=df['marks']>=50

df.to_csv('students_processed.csv',index=False)

print("Processed file saved. Preview:\n",df)
```

Output

```
Loaded data:
  name  marks class
0    A     78     I
1    B     45    II
```

```
2    C     65     I
3    D     38    II
```

Processed file saved. Preview:

```
   name  marks class   pass
0    A     78     I   True
1    B     45    II  False
2    C     65     I   True
3    D     38    II  False
```

Result

Pandas CSV file I/O was successfully performed by reading, processing, and writing student data. The pass column was correctly generated based on the marks.